

Business Objective

Problem Statement:

How can we characterize the structure and economic scale of the Texas livestock auction market, and which segments represent the greatest concentration of activity and openness to change?

Expected Business Outcome:

Objective 1: A segmentation framework that categorizes Texas livestock auction barns into 4 distinct market archetypes

Objective 2: Identification of which segments represent the greatest concentration of economic activity and which show the highest readiness for change.

Data and Exploratory Insights

Data Descriptions: 915 weekly USDA reports; 44 TX barns; Jan 2024 - Mar 2026

Cluster Visualization:



Feature Engineering

- **25 clustering features:** Median steer & heifer prices across 6 weight bands (200-800 lbs, Low & High), plus average head count per auction - all normalized to \$/cwt
- **Confidence weighting:** $w = \sqrt{(n_weeks)} / \max(\sqrt{(n_weeks)})$ - barns with more historical data weighted more heavily
- **Excluded features:** Sparse-data barns (<3 weeks) removed from clustering; retained only for anomaly detection
- **Standardization:** Z-score applied post-weighting so no single price band dominates

Methodology

Step	Method
Dimensionality Reduction	1. PCA (75% variance in 3 PCs) 2. MDS for geometric validation
Clustering	1. K-Means k=4, selected via Elbow + Silhouette; cross-validated with Agglomerative & DBSCAN
Anomaly Detection	1. Isolation Forest scored all 41 barns; flagged outliers treated separately

Cluster Result

Cluster	Avg Price (\$/cwt)	Avg Head/Auction
C1 - Premium South TX	~505	1,100+
C2 - Balanced Mid-Market	~391	~950
C3 - Below Avg Niche	~250	~700
C4 - East TX Piney Woods	~236	~900

Insights & Recommendations

- Market is Segmented, Not Uniform**
Four distinct archetypes exist with meaningful structural differences in price, volume, geography, and data quality. One-size-fits-all strategies will underperform.
- Value is Concentrated in C1 & C2**
C1 offers the highest per-transaction value; C2 offers the best combination of scale, stability, and accessibility. Together they represent the greatest concentration of economic activity.
- Readiness to Change Varies by Cluster**
C2 barns show the most operational consistency and data reliability (greatest readiness). C1 is high-value but deeply trust-based; requires careful, relationship-driven engagement.